



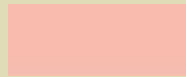
Sequence Diagram:

A sequence diagram is a type of Unified Modeling Language (UML) diagram used to represent the flow of messages or interactions between objects or components in a system. It depicts how objects communicate with each other over time to achieve a specific functionality.

Main Components of a Sequence Diagram

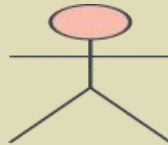
1. Object

Represents an instance of a class or an entity participating in the interaction. Each object is depicted as a rectangle with its name inside it.



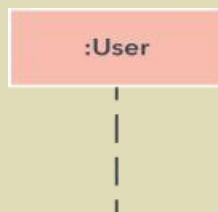
2. Actor

Represents an external entity that interacts with the system. It is typically depicted as a stick figure.



3. Lifeline

Represents the life of an object or actor during the interaction. It is shown as a vertical dashed line starting from the object or actor symbol.



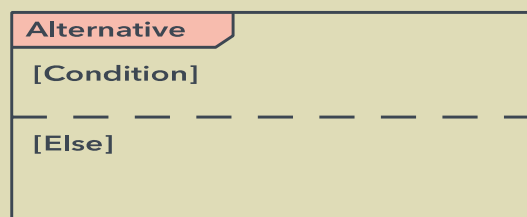
4. Activation Box

Represents the duration during which an object is performing an operation. It is shown as a rectangle on the lifeline.



5. Alternative (If/Else)

Represents conditional logic in the interaction. It is depicted using a frame labeled 'alt' with separate sections for each condition.



6. Loop

Represents a repetitive interaction. It is depicted using a frame labeled 'loop' with the iteration condition specified.



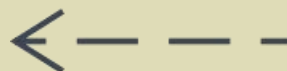
7-Message

Represents communication between objects. It is shown as a solid arrow pointing from the sender to the receiver.

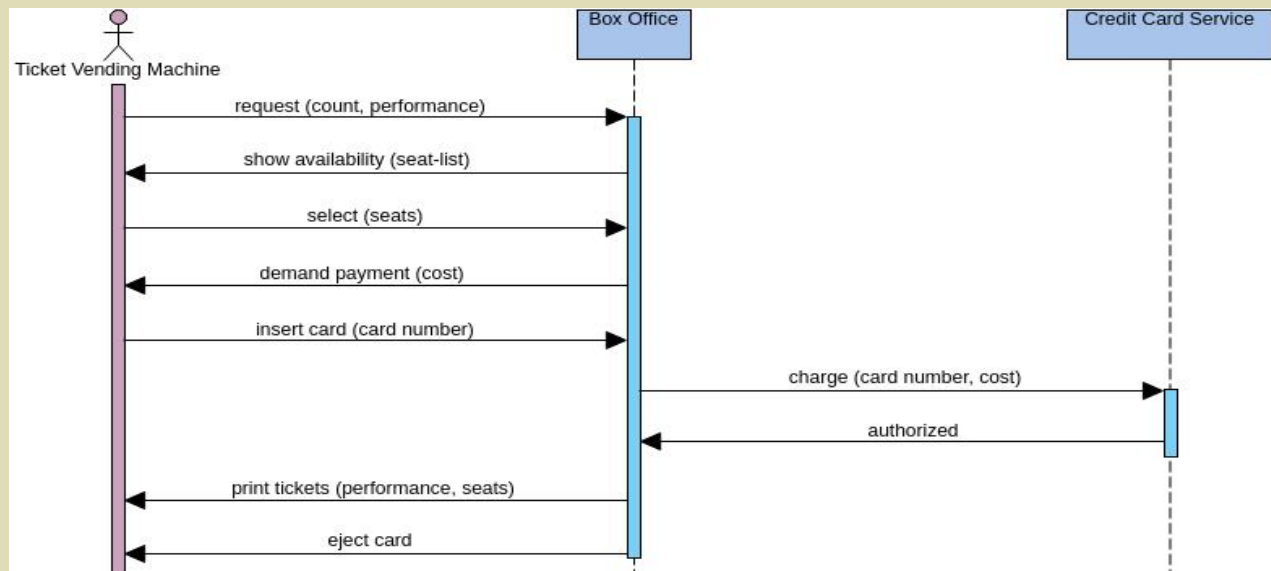


8. Return Message

Represents a response from the receiver to the sender. It is depicted as a dashed arrow pointing back to the sender.



EX1)



EX2)

